

Concurrency and Distributed Systems

Week 9 – Part 1: Python GIL Reality

Dr. Mohamad Aoude

Faculty of Engineering

May 13, 2026

Part 1 Roadmap

- 1 Runtime Limits
- 2 CPU-Bound Work
- 3 Demo

The Course Shift

Weeks 1–8

Java concurrency primitives, pools, futures, and debugging.

Week 9

The same concurrency vocabulary meets a different runtime.

Main warning

Concurrency API shape does not guarantee parallel CPU execution.

What the GIL Means Here

- CPython protects interpreter state with the Global Interpreter Lock.
- CPU-bound Python bytecode is constrained by that lock.
- Threads can still be useful for I/O-bound work.
- Workload matters: CPU-bound and I/O-bound are different claims.

CPU-Bound Loop

```
def cpu_bound_work(iterations):  
    total = 0  
    for i in range(iterations):  
        total += (i * i) % 97  
    return total
```

Question

What happens when several Python threads all run this loop?

Threads vs Cores

- More Python threads do not automatically mean more CPU cores doing Python bytecode work.
- Context switching may add overhead.
- The correct answer requires measurement and visual proof.

Run

```
.\run-demo.ps1 01
```

- Observe thread timing.
- Observe process timing.
- Do not overgeneralize from one small run.